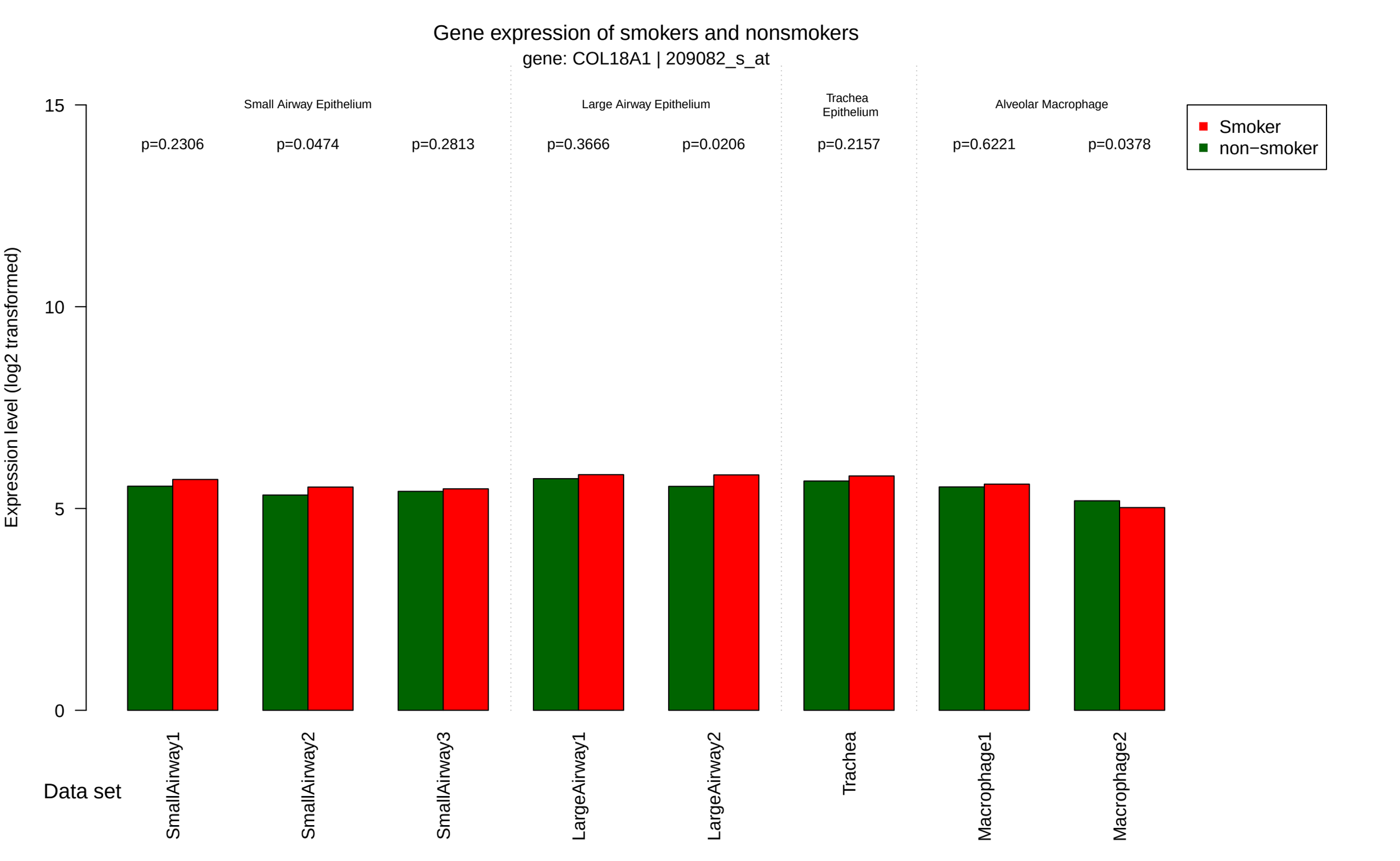
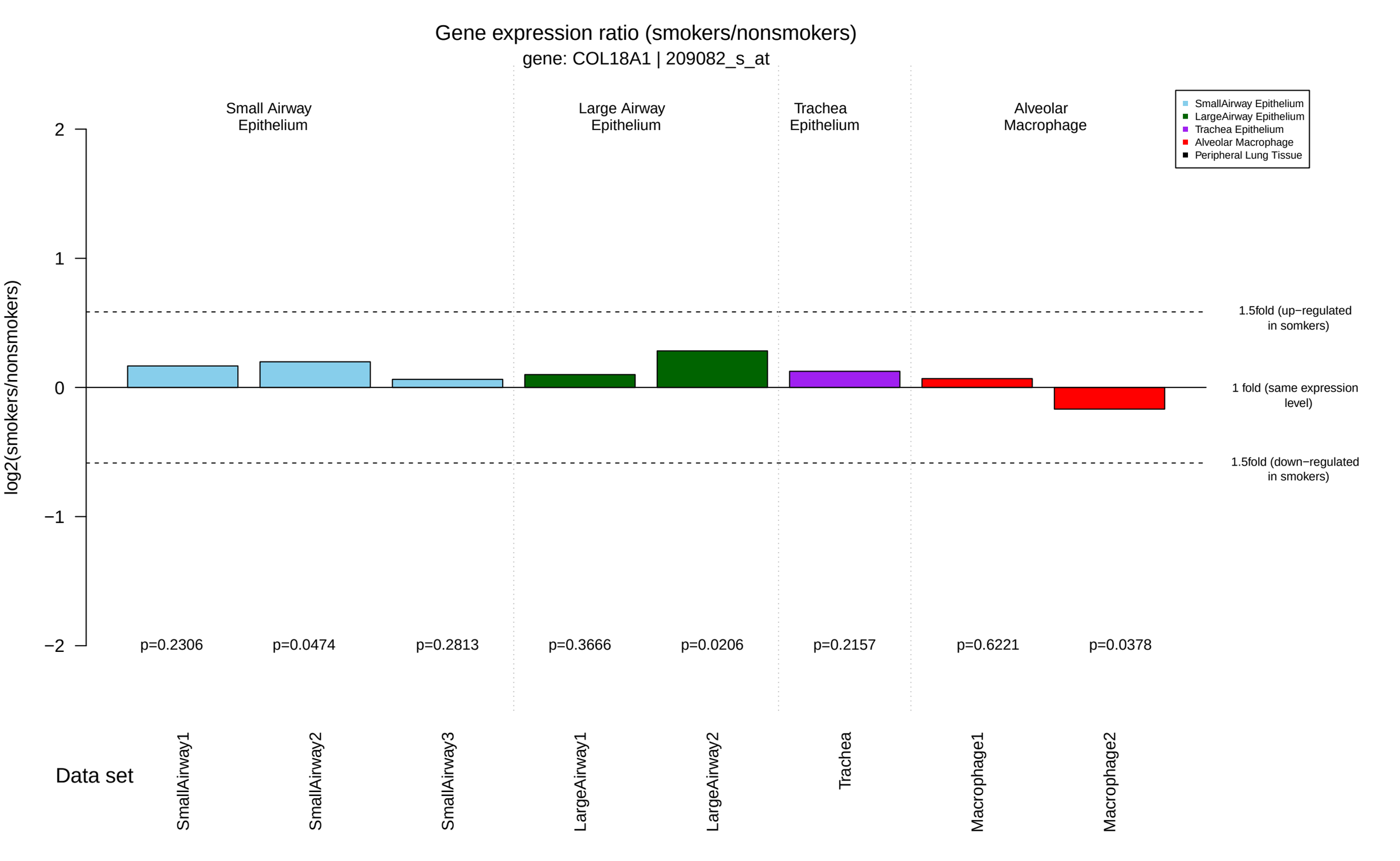
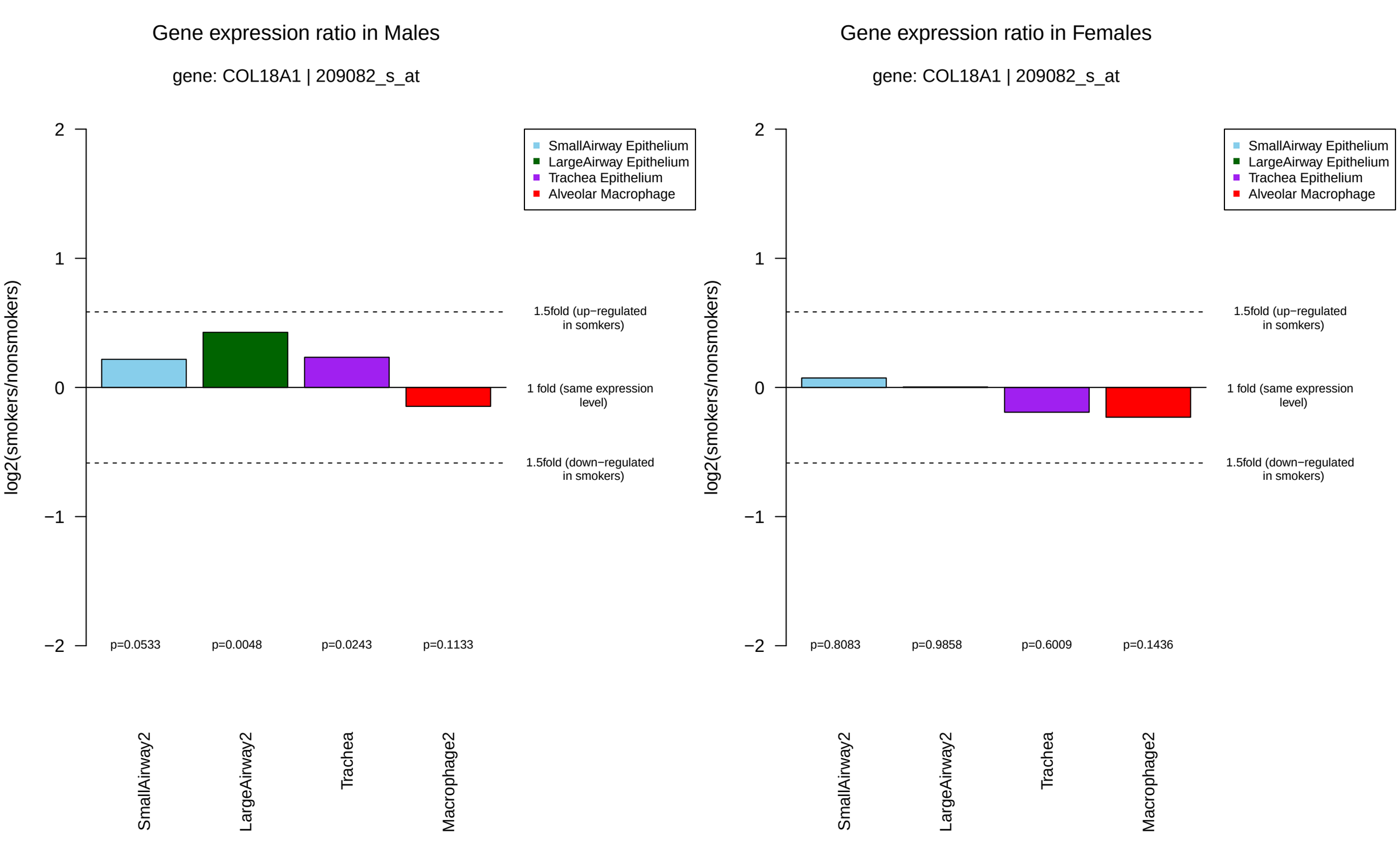
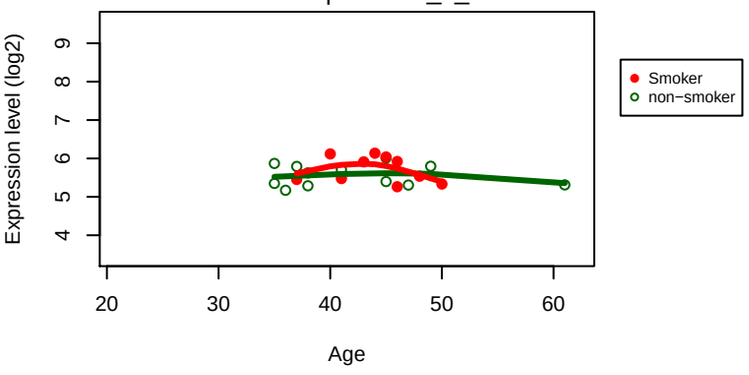
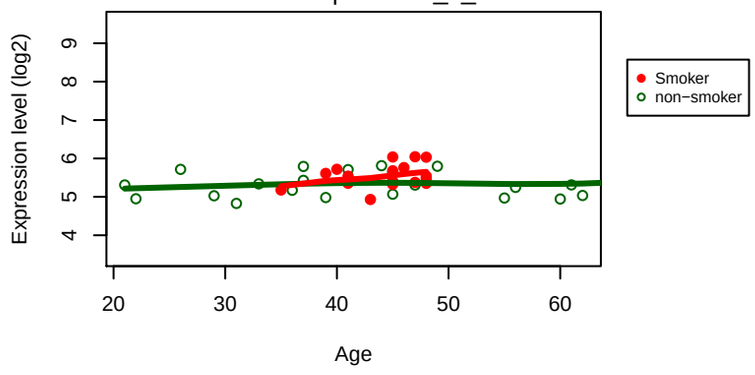
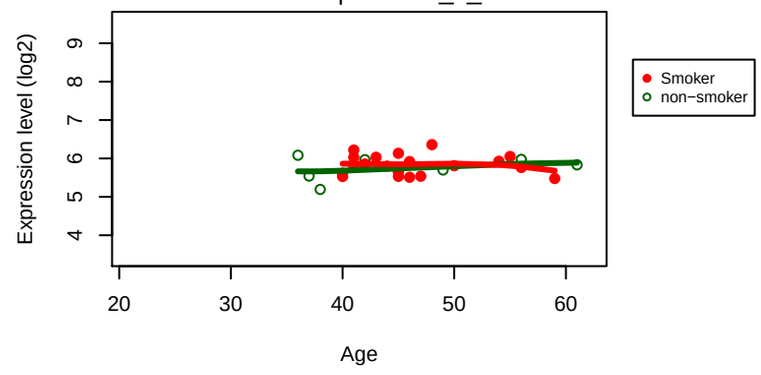

Gene expression of smokers and nonsmokers
gene: COL18A1 | 209082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2306
p=0.0474
p=0.2813
p=0.3666
p=0.0206
p=0.2157
p=0.6221
p=0.0378
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

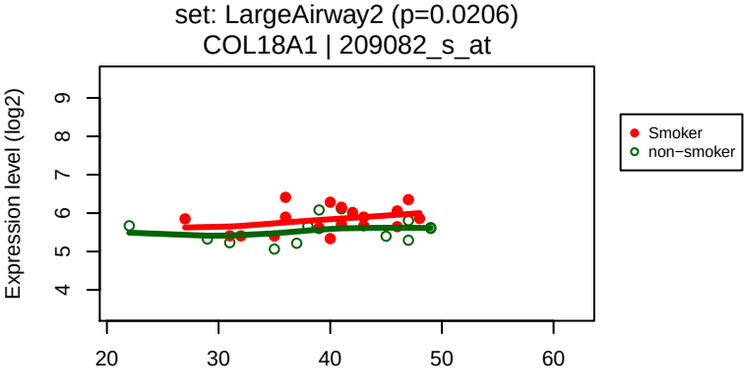

Gene expression ratio (smokers/nonsmokers)
gene: COL18A1 | 209082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2306
p=0.0474
p=0.2813
p=0.3666
p=0.0206
p=0.2157
p=0.6221
p=0.0378
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

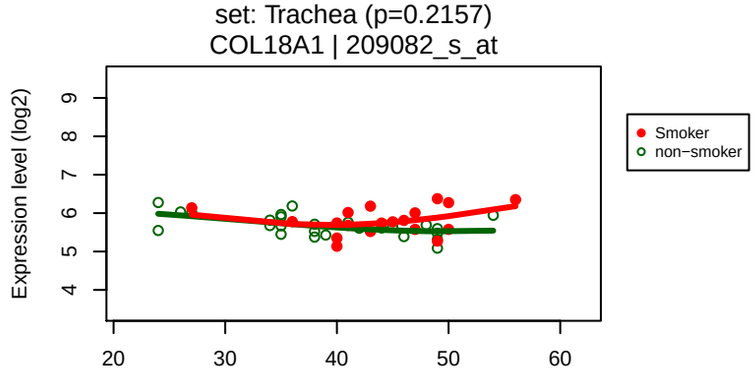

Gene expression ratio in Males
gene: COL18A1 | 209082_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0533
p=0.0048
p=0.0243
p=0.1133
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL18A1 | 209082_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8083
p=0.9858
p=0.6009
p=0.1436
SmallAirway2
LargeAirway2
Trachea
Macrophage2

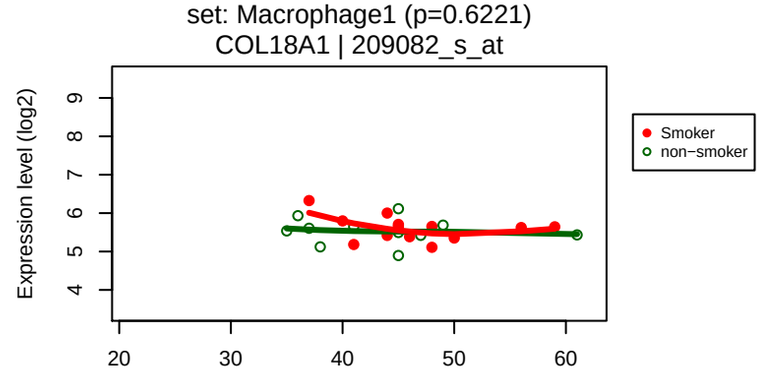

set: SmallAirway1 (p=0.2306)
COL18A1 | 209082_s_at
Expression level (log2)
Age
Smoker
non−smoker

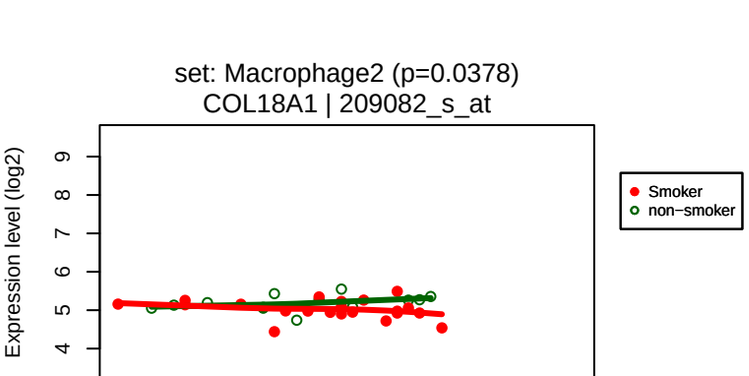

set: SmallAirway2 (p=0.0474)
COL18A1 | 209082_s_at
Expression level (log2)
Age
Smoker
non−smoker

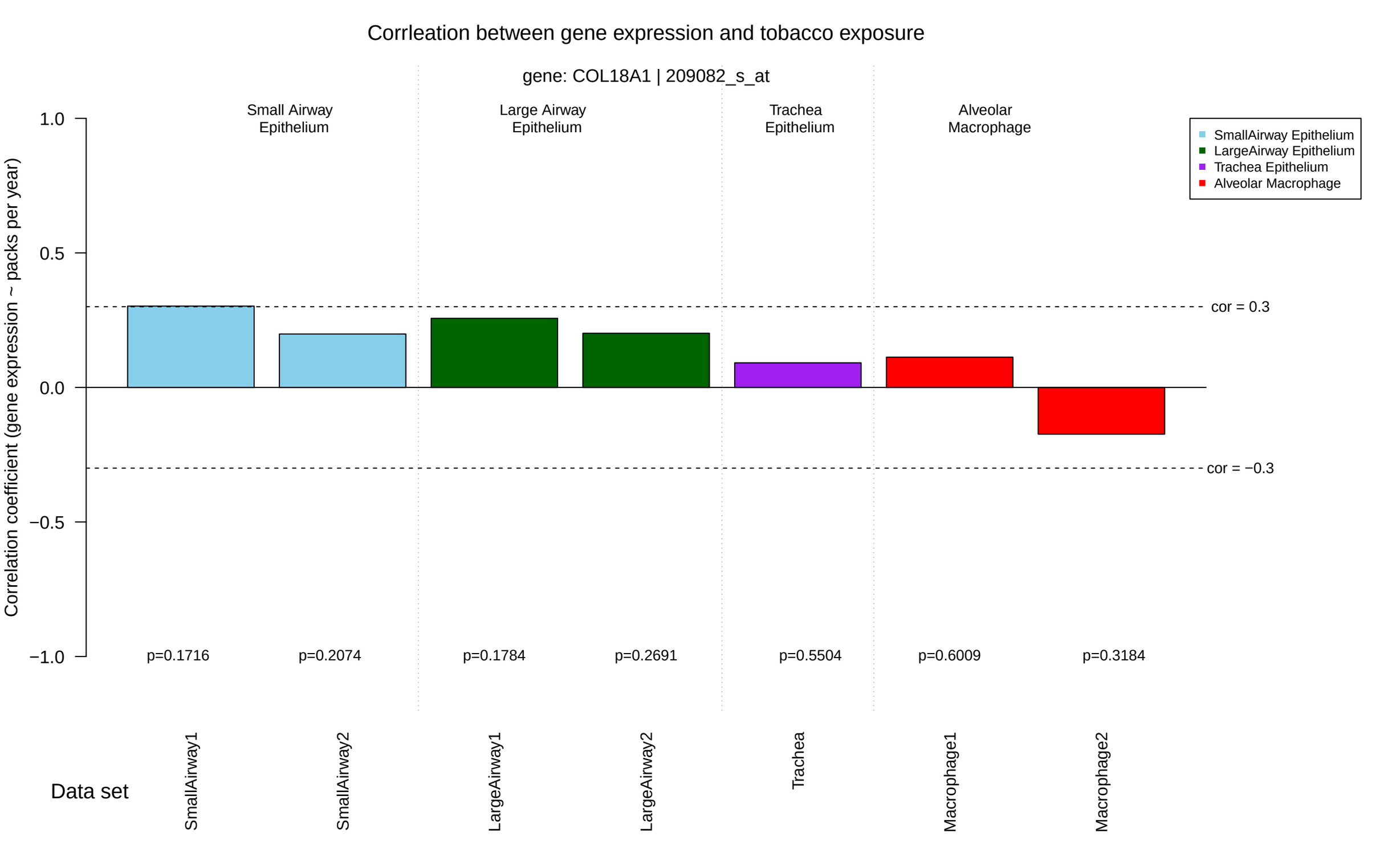

set: LargeAirway1 (p=0.3666)
COL18A1 | 209082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0206)
COL18A1 | 209082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2157)
COL18A1 | 209082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6221)
COL18A1 | 209082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0378)
COL18A1 | 209082_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL18A1 | 209082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1716
p=0.2074
p=0.1784
p=0.2691
p=0.5504
p=0.6009
p=0.3184
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2